

The Integration of Advanced NLP and Transformers for Multilingual AAC Devices

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Abstract

This study examines the benefits and limitations of Augmentative and Alternative Communication (AAC) devices, particularly high-technology applications, and explores how Natural Language Processing (NLP) techniques can support multilingual translation capabilities. AAC devices are widely used by people with communication disorders (PwCD), including people with Autism, Down syndrome, Amyotrophic Lateral Sclerosis (ALS), aphasia, and Parkinson's disease, to facilitate communication.

Although AAC technologies have expanded worldwide, particularly in North America and Europe, they continue to face significant barriers to accessibility. These barriers include high costs, limited insurance coverage, lack of personalization, and insufficient support for languages beyond English. Over a 14-week period, this research will evaluate AAC systems and their effectiveness in improving communication, identify gaps in current technologies, and investigate how NLP tools can improve translation functionality.

The findings will inform the development of more accessible and adaptive AAC technologies, with a particular focus on the use of artificial intelligence to promote cross-cultural and multilingual communication.

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1 Introduction

Augmentative and Alternative Communication (AAC) devices are used to enhance autonomy, communication, and decision-making for persons with communication disabilities, including those with Autism, Down Syndrome, Amyotrophic Lateral Sclerosis (ALS), aphasia, and Parkinson's disease. AAC devices fall within the broader

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domain of Assistive Technology (AT), including low-tech options such as communication boards and posters, as well as high-tech devices utilizing software and iOS applications[7] High-tech devices, particularly Speech-Generating Devices (SGDs), assist in the vocalization of PwCD, ranging from single utterances to full sentences.

However, numerous barriers limit the ability of PwCD to fully benefit from AAC devices. These barriers include limited translation capabilities in languages other than English, cognitive fatigue associated with prolonged use, high costs, insufficient insurance coverage, and broader societal perceptions of AAC use. Caregivers may not fully understand the necessity of AAC devices or may perceive them as reducing human connection in communication. Despite these challenges, AAC devices remain essential for promoting autonomy and communication among PwCD.

2 Related Work

Throughout this research, information has been gathered from multiple sources, including *Perceptions of AAC: An Ethnographic Investigation of Mexican-American Families*. This study highlights the barriers AAC devices present for Latinx families. Multiple caregivers reported a perceived lack of human connection and emotional intimacy when using AAC devices, viewing them as a hindrance to communication. Additionally, caregivers often receive limited instruction on device functionality, and vocabulary options are frequently only available in English. While PwCD can see the device as helpful to integrate their school and personal life, the mental exhaustion from use remains a common issue.

Information was also drawn from Daly's work, *ColorCode: A Bayesian Approach to Augmentative and Alternative Communication with Two Buttons*. This research explores the use of predictive querying to suggest words for persons with communication disabilities (PwCD). Daly proposes a novel AAC device designed for individuals with severe mobility limitations, utilizing a Bayesian framework. The system consists of two buttons—red and blue—which help predict the user's intended keyboard input through the ColorCode program. The findings indicate that ColorCode achieves a low error rate and requires an average of 2.07 clicks per character (cpc) to generate the correct symbol. By leveraging Bayesian logic and prior inputs, the system improves efficiency and reduces cognitive and physical load.

Additional research from *Disability and Rehabilitation: Assistive Technology* delves into the usage of an iOS application for AAC

devices. One key discrepancy in current mobile applications is the lack of individuality in word choice, limiting independence and communication for PwCD. These devices and their use for PwCD lack adequate monitoring with current insurance coverage, leading their beneficial aspects to be time-constrained. Lastly, these devices do not analyze or produce data on AAC usage in a trustworthy and efficient manner. However, mHealth applications are developing to mitigate or diminish these constraints.

The study evaluates the mobile application EuTalk, which produced generally positive results. The app includes training modules and game-like features to encourage user engagement, along with accessible data visualization through reports and graphics. The study involved four PwCD participants and 50 individuals without communication disabilities. The primary critiques included grammatical limitations and user fatigue. Notably, participants with more AAC experience reported fewer concerns regarding grammar. EuTalk also supports customization through user-defined buttons, promoting greater individuality in design.

3 Proposed Work

My proposed work consists of 14 weeks dedicated to exploring how NLP and Transformer models can generate multilingual language extensions for AAC devices, with the goal of reducing barriers faced by multilingual PwCD and their caregivers. At the end of this research period, I will produce a deliverable outlining how NLP techniques can support the adaptation of AAC devices, particularly through extensions that can be integrated into existing programs. This work aims to demonstrate how AAC systems can be updated to provide equitable communication access without stripping individuals of their native languages or disrupting familial communication.

Table 1: Fourteen-Week Research Timeline

Weeks	Focus
1–2	Survey existing AAC language systems and evaluate efficiencies
3–4	Identify gaps in the AAC market, especially multilingual support
5–8	Research NLP and Transformer-based translation extensions
9–11	Synthesize findings and outline next steps
12–14	Create research poster and finalize deliverable

4 Conclusion

In conclusion, AAC devices encompass a range of tools and methods designed to promote autonomy, communication, and connection for PwCD. However, barriers such as limited multilingual adaptation, high cost, insufficient insurance coverage, and lack of personalization continue to hinder their effectiveness for users and caregivers. My proposed research will examine the integration of NLP tools over a 14-week timeline to develop translation extensions for existing AAC systems. The final deliverable will outline key steps and templates that advance equitable, cross-cultural communication and accessibility for diverse linguistic communities.

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